WORKED SOLUTION

STEP I 2011, Question 10

Mechanics • Elastic Collisions

STEP I 2011, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q10

The key idea

Both particles start from rest at the same point, and nothing between release and the collision (free fall, a perfectly elastic bounce) loses any energy. So at any given height each particle has the same speed, and they meet at a common height. After that, the collision is a standard two-equation problem (momentum and Newton's law of restitution with $e = 1$), and the maximum height of B follows from energy once more.

Solution

We model A and B as particles, ignore air resistance, and take upwards as positive throughout.

Why the speeds are equal

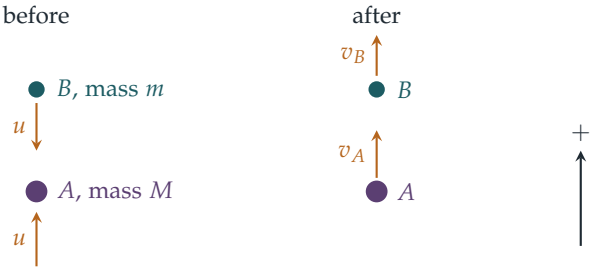
Each particle starts from rest at height h. While in free flight only gravity acts, so mechanical energy is conserved; the bounce of A on the plane is perfectly elastic, so it reverses A's velocity without changing its speed, and no energy is lost there either. Hence, per unit mass, each particle has total energy gh up to the first collision, and a particle at height y has speed v with

$$\tfrac{1}{2}v^2 + gy = gh, \qquad \text{so} \qquad v = \sqrt{2g(h-y)}.$$

The speed depends only on the height, not on the mass or on the particle's history. At the moment of the first collision the two particles are at the same height y, so their speeds are equal. Moreover A has not yet returned to P, so $y < h$ and this common speed $u = \sqrt{2g(h-y)}$ is positive. Since A has bounced it is moving up, and B (never having bounced) is moving down.

The collision

Immediately before the collision A has velocity $+u$ and B has velocity $-u$. Let their velocities immediately afterwards be v_A and v_B.



The impulsive forces in the collision are internal to the pair, and gravity gives no impulse over the instant of impact, so momentum is conserved:

$$Mu - mu = Mv_A + mv_B. \tag{1}$$

The collision is perfectly elastic ($e = 1$), so the speed of separation equals the speed of approach:

$$v_B - v_A = u - (-u) = 2u. \tag{2}$$

Substituting $v_B = v_A + 2u$ from (2) into (1):

$$(M-m)u = (M+m)v_A + 2mu \quad \Longrightarrow \quad v_A = \frac{(M-3m)u}{M+m},$$

and then

$$v_B = v_A + 2u = \frac{(M-3m)u + 2(M+m)u}{M+m} = \frac{(3M-m)u}{M+m}.$$

Since $M > 3m$ we have $M - 3m > 0$, so $v_A > 0$; and $v_B = v_A + 2u > 0$ too. So both particles move upwards after the first collision, as required.

Ask yourself How do we know B actually reaches the top of its flight before it next meets A?

After the collision both particles have acceleration $-g$, so their relative velocity stays $v_B - v_A = 2u$ while both are in flight: B pulls away from A at a steady rate, and they cannot meet before A next hits the plane. If A bounces while B is still rising, compare energies per unit mass: B has $\frac{1}{2}v_B^2 + gy$ and A has $\frac{1}{2}v_A^2 + gy$, which is smaller since $v_B > v_A > 0$ (and A's bounce keeps its energy). So whenever they are at the same height, A's speed is less than B's. To catch up with B from below while B is still rising, A would have to be rising at least as fast as B at the moment they meet, and we have just seen that it can't be. So the second collision happens only after B has passed its highest point, and that highest point is the maximum asked for.

Maximum height of B

The collision happens at height y where $u^2 = 2g(h-y)$, that is

$$y = h - \frac{u^2}{2g}.$$

After the collision B rises with initial speed v_B under gravity alone, so it climbs a further $v_B^2/(2g)$ (from $v^2 = v_B^2 - 2gs$ with $v = 0$). Its maximum height is therefore

$$\begin{aligned} y + \frac{v_B^2}{2g} &= h - \frac{u^2}{2g} + \frac{(3M-m)^2u^2}{2g(M+m)^2} \\ &= h + \frac{u^2}{2g(M+m)^2}\left[(3M-m)^2 - (M+m)^2\right] \\ &= h + \frac{u^2}{2g(M+m)^2}(2M-2m)(4M) \\ &= h + \frac{4M(M-m)u^2}{(M+m)^2 g}, \end{aligned}$$

using the difference of two squares in the third line. This is the required result.

Checks

- Energy: with v_A and v_B as above, $Mv_A^2 + mv_B^2 = (M+m)u^2$, as an elastic collision demands.
- Limiting case: if $m \to 0$, then $v_A \to u$ and $v_B \to 3u$ (a very light ball bounces off a heavy one as off a moving wall, leaving with velocity $2u$ relative to it), and the height becomes $h + 4u^2/g$, which matches $y + 9u^2/(2g)$.
- Since $M > m$, the extra height is positive: B ends up above P, as it should, having gained energy from A.
- Numerically, we released B at random times, with random h and masses satisfying $M > 3m$, stepped the motion (including A's further bounces) until the second collision, and found B's maximum height agreed with the formula to a relative error of about 10^{-10}.

Common slips

- Sign errors: taking both pre-collision velocities as $+u$ forgets that B is falling while A is rising.
- Measuring the extra climb $v_B^2/(2g)$ from the plane, or from P, rather than from the collision height $h - u^2/(2g)$.

Beyond STEP

The argument that speed depends only on height is energy conservation for a conservative force: the speed at a point depends only on the potential energy there, not on the route taken. The collision itself is often easiest in the centre-of-mass frame, where an elastic collision simply reverses each particle's velocity; transforming back gives v_A and v_B in one line. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

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